

NIL Youth Sports Platform — Budget Model

Version: 1.0 · **Date:** 2026-09-30 **Companion:** EXECUTION_PLAN.md (effort model §6), TEAM_BRIEF.md (§5.6 summary) **Audience:** whoever holds the budget — leadership, finance, or the founder

0. How to read this

This document turns the effort model in EXECUTION_PLAN.md §6 into money. It has three parts:

1. **§1–§3 — The model.** What costs what, and the parameters only you can supply.
2. **§4–§6 — The scenarios.** Three delivery shapes, with arithmetic, and an honest analysis of what AI agents actually change.
3. **§7–§11 — Run-rate, risk, guardrails, and what's excluded.**

Every vendor price in §2 was researched on 2026-09-30 and is dated and sourced. Prices change; verify before committing. Every *labor* rate is a **parameter you supply** — I have used one clearly-labelled illustrative rate so the arithmetic is visible, and all figures scale linearly if you substitute your own.

One number to internalise before reading further: an AI agent-week costs roughly **\$50–100**. A human engineer-week costs roughly **\$3,000**. Agents are 30–60× cheaper per unit of *produced* work — and this still only reduces the MVP build cost by ~25%, because the constraint is not agent spend. It is **human review capacity**, and the safety-critical work that agents must not own. §5 explains why.

1. Cost Model

Six components. Four are one-time build costs; two are ongoing.

#	Component	Type	Scale driver	Covered in
1	Human engineering labor	One-time	Engineer-weeks × loaded rate	§3, §4
2	AI agent capacity	One-time (during build)	Seats × months + metered usage	§3, §4
3	Legal & compliance review	One-time	External engagement	§3, §4
4	Infrastructure during build	One-time	Months of build	§4

#	Component	Type	Scale driver	Covered in
5	Infrastructure run-rate	Ongoing	Monthly, post-launch	\$7
6	Contingency	One-time	% of the above	\$8

Not modelled here (deliberately — see §11): revenue, Phase-4 finance infrastructure (escrow, KYC, payouts), post-MVP feature phases, marketing, and headcount beyond the build team.

2. Researched Reference Prices

Researched **2026-09-30**. Source type is flagged: **[P]** = vendor's own pricing page; **[S]** = third-party summary (treat as indicative, verify before committing).

2.1 Development infrastructure (during build)

Service	Plan	Price	Notes	Src
Supabase	Pro	\$25/mo + compute per project (~\$10/Micro, offset by \$10/mo compute credits)	Includes 100k MAU, 8 GB disk/project, 250 GB egress. Two projects (dev + prod) ≈ \$35–45/mo	[P]
Vercel	Pro	\$20/mo platform fee, includes 1 seat + \$20 usage credit; +\$20/mo per additional seat	3-person team ≈ \$60/mo	[P]
Railway	Pro	\$20/mo per workspace , unlimited seats, includes \$20 usage	Usage beyond: RAM \$10/GB/mo, CPU \$20/vCPU/mo, egress \$0.05/GB	[P]
Auth0	Free → B2C Essentials	\$0 up to 25,000 MAU; Essentials from \$35/mo (500 MAU) → \$350 (5,000) → \$700 (10,000)	B2C track applies (athletes/guardians/fans are consumers). B2B Essentials is \$150/mo+ if institution/sponsor orgs need it	[P]
Doppler	Team	<i>Not researched</i> — parameterise (~\$0–	Verify	—

Service	Plan	Price	Notes	Src
		30/mo)		
Domain + DNS	—	~\$10–20/yr	Edge-hosted, so DNS is free at the registrar	[P]

Build-period infra estimate: ~\$150–350/month, or ~\$900–2,500 over a 6-month build.

Cost-avoidance note: Auth0's free tier covers up to **25,000 MAU**. At MVP scale you may pay **\$0** for authentication. Verify against feature needs (custom domain, SSO, and role/org features may force a paid tier).

2.2 Production run-rate (post-launch)

Service	Plan	Price	Src
Sentry	Team	\$26/mo annual, \$29/mo monthly (Business \$80/\$89)	[P]
SendGrid	Essentials → Pro	from \$19.95/mo (50k emails) → from \$89.95/mo (100k)	[P]
OpenAI (RAG bot)	API	<i>Not researched</i> — parameterise. Usage-based; scales with bot traffic	—
CSAM / content-safety scanning	Vendor	Parameterise — required line item, not optional	—
Stripe	Per transaction	Not MVP-relevant — see §11	—

Post-launch run-rate: ~\$150–600/month at MVP scale (~\$1,800–7,200/year), excluding bot API usage and content-safety scanning.

The content-safety line item is the one people forget. TECHNICAL_SPEC.md §12.7 states that CSAM detection is legally mandated for any platform hosting minor imagery, even when media is third-party-hosted (thumbnails and metadata pass through). This is **not** a cost you can defer to Phase 4 — it is part of the P0 media moderation gate in Phase 2. Price it before you commit. Some CDN providers include a CSAM scanning tool at no extra cost; verify current terms rather than assuming.

2.3 AI agent capacity — the seat/usage split

Tool	Seat price	Billing shape	Src
Devin (Cognition)	Free \$0 · Pro \$20/mo · Max \$200/mo · Teams from \$80/mo (\$40/full seat) · Enterprise custom	Included quota, then on-demand credits at API list prices . (Legacy public per-ACU self-serve billing is retired; ACUs persist only in Enterprise order forms)	[P]
Cursor	Hobby \$0 · Pro \$20 · Pro+ \$60 · Ultra \$200 · Teams Standard \$40/user/mo , Premium \$120/user/mo	Seat includes a usage allowance, then on-demand at API rates	[P]
Claude Code	Free · Pro \$20 (\$17 annual) · Max 5x \$100 · Max 20x \$200 · Team standard \$25/seat/mo , premium \$125	Five-hour and weekly usage windows; Enterprise \$20/seat + usage at API rates	[P]

The budget-critical fact — seat price is a rounding error:

Anthropic publishes observed Claude Code costs across enterprise deployments of **~\$13 per developer per active day** and **\$150–250 per developer per month**, noting the seat fee is only **~8–12% of what you actually pay**. [S]

So: budget for usage, not seats. A \$20 seat can produce a \$250/month bill. For planning, use **\$200–400 per heavily-used agent seat per month, all-in** — which is **~\$50–100 per agent-week**.

2.4 Legal & compliance review

Not researchable generically — this is your single largest unknown. Review of a minors' platform across COPPA, FERPA, state high-school NIL divergence, and right-to-erasure is not commodity contract review. Parameterise at **\$15,000–60,000**, and treat the high end as live until you have a quote.

This is decision **D10** in `EXECUTION_PLAN.md` §1 — the longest lead time in the project, and it hard-gates the Stage 2 schema freeze.

3. Parameters You Must Supply

Substitute your own values. Every scenario below scales linearly.

#	Parameter	Symbol	Illustrative value used here	Your value
1	Fully-loaded engineer cost per week (salary × ~1.3 + overhead ÷ 46 working weeks)	R	\$3,000	_____
2	Senior/lead rate (for the safety-core owner)	R _s	\$4,000	_____
3	Number of human engineers	H	2–3	_____
4	Agent seats	A	3	_____
5	Agent all-in cost per seat-month	A _m	\$300	_____
6	Agent effectiveness multiplier on verifiable work	E	2× (range 1.5–3×)	_____
7	Review tax — human hours per unit of agent output	T	25%	_____
8	Legal review	L	\$30,000 (range \$15–60k)	_____
9	Build duration in months	M	7	_____

Parameter 6 is the one to measure, not guess. The S0.1 extraction spike (EXECUTION_PLAN.md §0) already deletes one commerce vertical and repairs it to a green typecheck. **Have the agent do that work, and record the agent-hours, the human review-hours, and the defects found.** That single measurement replaces parameters 6 and 7 with data — and it is the same 3–5 days the plan already recommends.

4. Scenarios

4.1 The work split (from EXECUTION_PLAN.md §6.1, mid estimates)

The 56 ew midpoint divides into work agents can safely produce, and work they must not own.

Stage	Mid ew	Class	Why
S0 Baseline, spike, CI, hygiene	2.0	Human-led	Decisions and measurement
S1 Infrastructure	1.5	Human-led	Account setup, credentials, auth wiring
S2 Schema Foundation	6.0	Mixed	Schema design is human; migrations, seeds, and table definitions are agent-heavy
S3 API Extraction & Core	9.0	Agent-heavy	Deletion-and-repair loops, 19 ID generators, 11 resolvers — all verifiable by typecheck
S4 Safety Substrate	3.0	Human-owned	Tenant guard, consent engine, moderation gate, test harness
S5 Vertical Slice	3.5	Human-owned	The five P0 invariants
S6 Web Extraction & Core	9.0	Agent-heavy	Same repair loop, verifiable by typecheck
S7 Credibility Shell	2.5	Mixed	UI is agent-heavy; the lead-intake safety path is human
S8 Native Pipeline Breadth	3.5	Mixed	Entity types agent-heavy; the compliance gate is human
S9 Unified Service	16.0	Mixed	Capabilities/dashboards agent-heavy; invitations, CRM, and bot guardrails human
Total	56.0		

Resulting split: ~34 ew agent-suitable, ~22 ew human-owned.

4.2 Scenario A — All-human (3 engineers)

Line	Calculation	Mid	Range
Engineering labor	56 ew × \$3,000	\$168,000	\$118,500–\$219,000
Agent capacity	—	\$0	\$0
Legal review	parameter	\$30,000	\$15,000–\$60,000
Build infra	7 months × \$250	\$1,750	\$900–\$2,500

Line	Calculation	Mid	Range
Total		\$199,750	\$134,400–\$281,500

Highest cost, lowest execution risk, no new tooling to learn.

4.3 Scenario B — Hybrid (2 humans + 3 agent seats) — recommended

The agent takes the 34 ew of verifiable work at 2× effectiveness; humans own the 22 ew safety core and review the agent's output.

Line	Calculation	Mid	Range
Agent-produced work	34 ew ÷ 2×	17 ew	11–23 ew
Human review of it	17 ew × 25%	4.3 ew	3–6 ew
Human-owned work	—	22 ew	22 ew
Total human		43.3 ew	36–51 ew
Engineering labor	43.3 ew × \$3,000	\$129,900	\$108,000–\$153,000
Agent capacity	3 seats × 7 mo × \$300	\$6,300	\$4,200–\$8,400
Legal review	parameter	\$30,000	\$15,000–\$60,000
Build infra	7 months × \$250	\$1,750	\$900–\$2,500
Total		\$167,950	\$128,100–\$223,900

Saving vs Scenario A: ~16% on the total, ~23% on labor. The saving is smaller than the agent cost advantage suggests — §5 explains why.

4.4 Scenario C — Agent-heavy (1 human + 5 agent seats)

Line	Calculation	Mid	Range
Human-owned work	—	22 ew	22 ew
Human review of agent output	—	<i>unbounded — see below</i>	—
Engineering labor	22 ew × \$4,000 (senior, as sole owner)	\$88,000	\$88,000
Agent capacity	5 seats × 9 mo × \$300	\$13,500	—
Legal review	parameter	\$30,000	\$15,000–\$60,000
Build infra	9 months × \$250	\$2,250	—
Total (on paper)		\$133,750	—

Cheapest on paper, most expensive in expectation. Not recommended.

One human cannot own 22 ew of safety-critical design *and* adversarially review 34 ew of agent output. The P0 gates specifically require a second pair of eyes — a tenant guard reviewed by its own author is not reviewed. This scenario does not reduce cost; it **moves cost from the build budget into the incident budget**, which is the one line item you cannot bound.

The calendar also stretches rather than compresses, because a single reviewer is a serial resource.

4.5 Scenario comparison

	A — All-human	B — Hybrid	C — Agent-heavy
Team	3 humans	2 humans + 3 agents	1 human + 5 agents
Mid total	\$199,750	\$167,950	\$133,750
Range	\$134k–282k	\$128k–224k	not bounded
Calendar	5–8 months	5–8 months	9+ months
Independent review of the safety core	☒ 3 reviewers	☒ 2 reviewers	✗ none
Recommendation	Viable, highest cost	Recommended	Not for a minors' platform

5. What AI Agents Actually Change

This is the section worth reading twice, because the honest answer is not the intuitive one.

5.1 The economics are dramatic per unit of work

	Cost per week
Human engineer	~\$3,000
AI agent (heavily used, all-in)	~\$50–100

Agents are **30–60× cheaper** per unit of *produced* work. If agents were 2× effective on the whole project, the build would cost a small fraction of Scenario A.

5.2 ...and it still only saves ~25%, for three structural reasons

1. **The safety core cannot be delegated.** ~22 ew — the tenant guard, consent engine, visibility firewall, moderation gate, cache-invalidation contract, and the P0 test suite — must be human-designed and human-reviewed. Not because agents can't write the code, but because in a platform handling minors' data, the failure mode is a legal event, and the plan's whole P0 regime assumes an accountable author.
2. **The review tax is real and higher here.** Agent output is cheap to produce and expensive to

verify. The plan's own standard (EXECUTION_PLAN.md §7.1) requires every negative-path test to be **non-vacuous** — it must fail if the control is removed. Verifying that an agent's safety test genuinely tests the safety property is human work, and it does not compress.

3. **The critical path is decisions and review, not typing.** $S2 \rightarrow S3 \rightarrow S4 \rightarrow S5$ is a hard chain (EXECUTION_PLAN.md §2.1). Agents do not shorten it, because each stage gates on a human decision (D1–D5) or a human review.

5.3 Where agents genuinely earn their keep on this project

Ranked by value, based on the stage classification in §4.1:

Rank	Work	Why it suits agents
1	S3 / S6 deletion-and-repair (~18 ew)	Mechanical, bounded, and verifiable by a green typecheck — the cheapest possible verification signal
2	Repetitive analog work — 11 resolvers (optimization O2), 7 base singletons, ~44 components, 19 ID generators, seed scripts, nav reseed	One reference implementation, then N near-copies
3	Spec reconciliation and documentation	Exactly what produced SPEC_AMENDMENTS.md — high-volume, low-risk, reviewable by diff
4	Test scaffolding	Fast to generate, cheap to check
5	Schema migrations and seeds	Pattern-following; verified by <code>prisma validate</code>

5.4 Where agents must not be the owner

Work	Why
The five P0 invariants (S5.7)	A defect is a legal event. Requires human design <i>and</i> independent human review
The tenant guard (S4.1)	It is the Phase-2 substitute for database-level isolation (finding F1). Its correctness is the privacy guarantee
The public-surface PII sweep (S4.6)	The failure mode is <i>something missing</i> — and agents are systematically biased toward confirming what they were asked to look for. This needs human adversarial review
Decisions D1–D7	Architecture and policy. Not delegable

Work	Why
Legal sign-off (D10)	Not delegable at any price

5.5 The metric to manage

From the Devin pricing analysis, and worth adopting verbatim as the budget KPI:

Price agent work by accepted work per hour of human review — not by prompts, sessions, PRs, or runtime.

A PR count tells you nothing. "We spent 4 human-hours to land 30 ew-equivalent of verified change" tells you everything. Track it per stage; if it degrades, the multiplier E in §3 is wrong and the estimate needs revising.

5.6 On the sensitivity of this question

Worth stating plainly, because it is a fair thing for a team to wonder about.

This model does **not** propose headcount reduction. Scenario B is the recommended shape, and it keeps **two** human engineers — including a dedicated owner for the safety core. What agents change is *where human attention is spent*: away from mechanical breadth work, toward the safety-critical design, review, and decisions that determine whether the platform is defensible.

Scenario C — the one that does reduce headcount — is explicitly **not recommended**, on technical grounds rather than sentiment: it removes independent review from the child-safety controls, which is the single most expensive risk in the project (§8). Anyone proposing it is either over-estimating what agents can own or under-budgeting review.

6. Cost by Milestone (Scenario B)

The natural budget checkpoints, since each milestone already has a gate and an evidence artifact.

Milestone	Cumulative effort	Human-equivalent (~77%)	Cumulative cost	Cumulative time
M0 — Foundations	18.5 ew	~14.2 ew	~\$43,000	~2 months
M1 — Investor-ready public site	21 ew	~16.2 ew	~\$49,000	~3 months
M2 — Safety core validated ★	27.5 ew	~21.2 ew	~\$64,000	~3.5–4 months
M3 — Platform complete	40 ew	~30.8 ew	~\$92,000	~5 months

Milestone	Cumulative effort	Human-equivalent (~77%)	Cumulative cost	Cumulative time
M4 — Full MVP	56 ew	~43.3 ew	~\$130,000 (labor)	5–8 months
Add: agent capacity, infra, legal	—	—	+\$38,000	—
Program total			~\$168,000	5–8 months

M2 is the budget decision point. It is where the architecture is proven and where EXECUTION_PLAN.md §6.5 re-baselines the remaining stages from measured velocity. Approve the budget through M2 with confidence; treat M3–M4 as provisional until then.

7. Run-Rate After Launch

Line	Monthly	Notes
Supabase Pro (+1 extra project)	\$35–45	Compute per project
Vercel Pro (3 seats)	\$60	\$20 platform + 2 × \$20 seats
Railway Pro	\$20–40	Scales with API load
Auth0	\$0–350	Free to 25k MAU; B2C Essentials \$35 @500 MAU → \$350 @5,000
Sentry Team	\$26–29	Annual vs monthly billing
SendGrid Essentials	\$20	50k emails
OpenAI (RAG bot)	<i>parameterise</i>	Scales with bot traffic
Content-safety / CSAM scanning	<i>parameterise</i>	Required, not optional — see §2.2
Doppler	<i>parameterise</i>	~\$0–30
Domain	~\$1–2	Annual, amortised
Total	~\$150–600/mo	~\$1,800–7,200/yr

Post-MVP, Phase 4 adds real cost: escrow and payment processing, KYC/AML verification per guardian, tax-document (1099) handling, and the non-profit allocation ledger. **None of that is in this budget.** It is scoped in EXECUTION_PLAN.md §4 Stage 10 (outline only, no estimate).

7.1 The revenue gap — set this expectation explicitly

The MVP generates no revenue. Implementation Phase 1–3 delivers a public presence, a controlled

profile pipeline, and persona portals. The monetisation model (TECHNICAL_SPEC.md §12.10 — transaction fees on settled deals, plus payer-keyed institution/sponsor tiers) is **Phase 4**.

So the MVP is a pure cost centre. It is funded by the credibility it creates, not by what it collects. Anyone building a budget model that assumes the MVP pays for itself is modelling Phase 4.

8. Contingency, and the Cost of Failure

8.1 Contingency

Estimate portion	Confidence	Recommended contingency
S0, S1, S4, S5, S7 (well-understood)	High/Medium	10%
S2, S3, S6, S9 (Low confidence)	Low	25%
Legal review	Unknown	Hold the full \$60,000 until quoted
Agent capacity	Usage-metered	Cap it — see §9

Blended recommendation: 15–20% on the mid estimate, i.e. ~\$25,000–35,000 on Scenario B.

8.2 The cost of failure

The reason the safety budget is not compressible:

Failure	Rough cost shape
A minor's data reaches a public surface	Legal exposure, mandatory notification, remediation, and reputational loss in a community where trust is the product . Not boundable in advance
A consent revocation that doesn't fully evict	Same, plus a demonstrable control failure — the plan's own P0 test 3 exists precisely for this
An adult-to-minor contact path	The single most damaging possible outcome for this platform
Erasure that can't complete because the audit log holds PII	Regulatory exposure, and a data migration to fix (finding F9 / decision D7)

Every one of these is cheaper to prevent than to remediate. The ~\$25–35k of contingency and the ~22 ew of human-owned safety work are the cheapest insurance in this budget. Trading either for a lower headline number is a false economy — and it is the specific failure mode that Scenario C represents.

9. Budget Guardrails

Practical controls, so the budget behaves:

#	Guardrail	Why
1	Cap agent on-demand spend monthly per seat, with an alert threshold	Usage is metered at API list prices; a runaway loop is a real bill. All three tools support spend limits
2	Track accepted-work-per-review-hour (\$5.5) per stage	The only meaningful agent productivity metric; it validates or refutes parameter E
3	No agent output merges to a P0 surface without human review	Non-negotiable. Make it a branch-protection rule, not a convention
4	Re-baseline the budget at M2 alongside the effort re-baseline	M3–M4 are provisional until the architecture is proven
5	Hold legal contingency unspent until quoted	The largest single unknown; do not absorb it into other lines
6	Use the free tiers deliberately — Auth0 to 25k MAU, Supabase/Vercel/Railway free tiers during exploration	Meaningful at MVP scale; revisit when limits bind
7	Price content-safety scanning before committing to launch	Required, not optional, and easy to discover late

10. Sensitivity — What Moves the Number

Driver	Effect on Scenario B mid (\$168k)
Legal review at \$15k instead of \$30k	−\$15,000 (−9%)
Legal review at \$60k instead of \$30k	+\$30,000 (+18%) — the largest single swing
Agent multiplier E 3× instead of 2×	−\$21,000 (−13%)
Agent multiplier E 1.5× instead of 2×	+\$21,000 (+13%)
Review tax T 40% instead of 25%	+\$7,500 (+4%)
Labor rate ±20%	±\$26,000 (±15%)
Effort at the low end (39.5 ew)	−\$42,000 (−25%)
Effort at the high end (73 ew)	+\$42,000 (+25%)

Ranked by leverage: (1) effort range, (2) legal review, (3) labor rate, (4) agent multiplier, (5) review tax.

Note the asymmetry worth exploiting: the **agent multiplier and the review tax are the two drivers that move together**. A higher E (agents more effective) and a lower T (cheaper to verify) compound — which is exactly what optimization O2 (template-first capability) and the CI gates in S0.3 are for. Verifiable-by-construction work is where agents win twice: once in production, once in review.

The first two are the ones to attack. Effort is attacked by the S0.1 spike (which measures the repair tax). Legal is attacked by getting a quote early — which is also decision D10, the plan's longest lead-time item.

11. What Is Not In This Budget

Excluded	Why	Where it lives
Revenue	MVP is a cost centre (§7.1)	Phase 4, TECHNICAL_SPEC.md §12.10
Payment processing / Stripe fees	No money moves in the MVP	Phase 4
Escrow, KYC/AML, payouts, 1099 handling	Explicitly post-MVP (EXECUTION_PLAN.md §4 Stage 10)	Phase 4 — material cost, unestimated
Automated compliance vetting	Manual verdict in MVP	Phase 4
Row-level security implementation	Decision D1 default = Phase 4 (+3–6 ew if elected early)	EXECUTION_PLAN.md §1 D1
Data-erasure automation	Manual in MVP	Phase 4
Marketing, brand, content production	Outside engineering scope	Separate budget
Post-launch support and on-call	Not modelled	Separate budget
Headcount beyond the build team	Not modelled	Separate budget
Manual compliance-review staffing	A launch decision, not an engineering one — MVP launch volume is bounded by human review capacity (TEAM_BRIEF.md §7)	Separate budget, needed before launch

12. Summary

Question	Answer
MVP build cost, recommended (Scenario B)	~\$168,000 mid · range \$128,000–224,000
Same, all-human (Scenario A)	~\$200,000 mid · range \$134,000–282,000
Saving from AI agents	~16% of total, ~23% of labor — not an order of magnitude (§5.2)
Agent cost per seat-month	\$200–400 all-in (budget usage, not seats — §2.3)
Biggest single line	Human engineering labor , ~77% of the total
Biggest single unknown	Legal review , \$15k–60k — get a quote early
Post-launch run-rate	~\$150–600/month
Contingency	15–20%, weighted to the Low-confidence stages
When the budget becomes trustworthy	At M2 , alongside the effort re-baseline
The one thing not to compress	The ~22 ew of human-owned safety work and its independent review (§8.2)
The one measurement to take first	Agent effectiveness and review cost, during the S0.1 spike (§3, parameter 6)

Bottom line. The MVP costs roughly **\$170k** in the recommended shape, and **~\$130k of that is human engineering** — because over a third of the work is safety-critical and cannot be delegated, and because the rest still carries a human review tax. AI agents are genuinely cheap per unit of produced work (~~\$50–100/agent-week~~ vs \$3,000/engineer-week) and are worth deploying aggressively on the mechanical breadth work, which is where the plan's uncertainty is concentrated. But they reduce the MVP bill by about a quarter, not by a factor of ten — and the scenarios that promise more do so by removing independent review from child-safety controls, which transfers cost into the one budget you cannot bound.